

Wide Bandgap Perovskites: A Comprehensive Review of Recent Developments and Innovations

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Recent advances in wide-bandgap (WBG) perovskite solar cells (PSCs) demonstrate a burgeoning potential to significantly enhance photovoltaic efficiencies beyond the Shockley–Queisser limit for single-junction cells. This review explores the multifaceted improvements in WBG PSCs, focusing on novel compositions, halide substitution strategies, and innovative device architectures. The substitution of iodine with bromine and organic ions such as FA and MA with Cs in the perovskite lattice is emphasized for its effectiveness in achieving higher open-circuit voltages and reduced thermalization losses. Furthermore, the integration of advanced charge transport layers and interface engineering techniques is discussed as critical to minimizing open-circuit voltage (V_{OC}) deficits and improving the photo-stability of these cells. The utilization of WBG PSCs in diverse applications such as semitransparent devices, indoor photovoltaics, and multijunction tandem devices is also explored, addressing both their current limitations and potential solutions. The review culminates in a comprehensive assessment of the current challenges impeding the industrial scale-up of WBG PSC technology and offers a perspective on future research directions aimed at realizing highly efficient and stable WBG PSCs for commercial photovoltaic applications.

light-emitting diodes (LEDs) and even laser devices.^[10–13] A pivotal attribute of 3D perovskites is their customizable bandgap, which can be precisely adjusted by substituting different elements within the perovskite crystal structure. This flexibility allows for the engineering of materials specifically optimized for various functional requirements, thereby enhancing their efficiency and broadening their potential applications.^[14–16]

Wide bandgap (WBG) perovskites are characterized by their larger bandgap energies, offering several substantial advantages for semiconductor devices. Typically, wide bandgap perovskites exhibit bandgap energies ranging between 1.7 and 2.4 eV, which allows them to absorb high-energy photons efficiently while reducing thermalization losses. This bandgap range is particularly suitable for applications that require high photovoltage and selective photon utilization.^[17] WBG perovskites are especially effective in minimizing thermalization losses, a common inefficiency where photons lose energy as heat before

conversion to electricity.^[18,19] By optimizing the bandgap, WBG perovskites facilitate higher open-circuit voltages and enable more effective utilization of the solar spectrum.^[20,21] This capability potentially allows these materials to surpass the performance constraints imposed by the traditional Shockley–Queisser limit for single-junction solar cells. Moreover, the stability of WBG perovskites underpins their utility, particularly for the longevity and durability of solar cells. Their robustness against environmental factors such as moisture, heat, and ultraviolet radiation makes them suitable for outdoor applications, where durable materials are essential.^[22–25] In addition to their advantageous bandgap range, WBG perovskites possess unique physical properties that expand their application potential. Specifically, their large exciton binding energies and deeper valence band levels make them more favorable for selecting *p*-type materials, thereby broadening design flexibility in device architecture.^[26–28] Furthermore, WBG perovskites enable the development of innovative devices like semitransparent solar cells, multijunction tandem cells, and light-emitting diodes (LEDs). Their ability to selectively absorb or emit photons optimizes energy conversion efficiency, making them highly versatile for advanced optoelectronic applications.^[29–37]

1. Introduction

The advancement of photovoltaic and optoelectronic technologies has been significantly propelled by the development of perovskite materials, which are renowned for their exceptional optoelectronic properties and adaptability.^[1–6] Among these, 3D perovskites are distinguished by their high absorption coefficients, tunable bandgaps, and straightforward manufacturing processes.^[7–9] These characteristics render them highly suitable for a broad array of applications, ranging from solar cells to

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Achieving a wide bandgap in perovskites often involves strategic modifications to the material's composition. A common approach is the substitution of halides-replacing iodine with bromine or chlorine to effectively increase the bandgap.^[38–40] Additionally, altering the cationic components of the perovskite structure, such as replacing organic cations like formamidinium (FA) and methylammonium (MA) with inorganic cesium (Cs) not only adjusts the optical properties but also enhances the stability of the materials.^[41–43] The evolution from traditional 3D structures to layered 2D perovskites is another strategic approach for increasing both bandgap and stability.^[44,45] In 2D structures, organic spacer cations are interspersed between layers of inorganic perovskite materials, inducing quantum and dielectric confinement effects.^[46–48] These effects not only broaden the bandgap but also significantly improve the material's environmental stability, thereby expanding their practical applications. Current research also focuses on exploring non-perovskite and perovskite-like materials, such as double perovskites or rudorffite, which avoid using toxic lead by substituting it with environmentally friendlier elements like tin or germanium.^[49–51] These efforts aim to address environmental and safety concerns associated with traditional perovskite materials and open new avenues for tailoring electronic and optical properties to enhance performance.

Despite the promising prospects of WBG perovskites, they face several challenges that need to be addressed to advance their commercial viability. These challenges include phase stability, ion migration, and long-term degradation under operational conditions.^[22] Researchers are increasingly focused on improving material formulations, developing advanced encapsulation techniques, and engineering novel interfaces and charge transport layers to enhance the operational stability and efficiency of perovskite devices.^[52–54] This review explores advancements in WBG perovskite solar cells (PSCs) with bandgaps of 1.7 eV or greater. The review focuses on material innovations, halide engineering, new device architectures, and advanced charge transport layers. Also, various applications in tandem solar cells and indoor photovoltaics are discussed. Key challenges like phase stability and ion migration are discussed, with future research directions aimed at achieving efficient, stable, and commercially viable WBG PSCs.

2. Halide Engineering in Wide Bandgap Perovskite

The size of halide ions plays a crucial role in determining the crystal lattice parameters and overall stability of the perovskite structure. Iodine, being larger than bromine and chlorine, leads to a larger lattice constant. Substituting iodine with smaller halide ions decreases the lattice constant, enhancing structural stability and altering the material's electronic properties.^[39,55,56] This substitution is essential for tuning the bandgap, which is crucial for increasing the open-circuit voltage (V_{OC}) and making the material suitable for applications like tandem solar cells that require higher V_{OC} . Substituting iodine with bromine in methylammonium lead iodide (MAPbI₃) increases the bandgap due to bromine's smaller ionic radius. This change leads to a higher degree of orbital overlap, increasing the energy difference between the valence band maximum (VBM) and the conduction band minimum (CBM).^[57,58] The resulting bandgap widening shifts

the absorption edge toward shorter wavelengths, enhancing the absorption of high-energy photons and reducing thermalization losses. Furthermore, bromine substitution enhances the thermal stability of perovskite films by creating a more tightly packed lattice, which is less prone to thermal degradation.^[59,60] The ability to tune the bandgap by varying the halide composition is a pivotal feature of perovskite materials. This tuning is controlled by the electronic structure of the halides and their interaction with lead (Pb) and the organic cation, such as MA or FA. The conduction band primarily comprises Pb *p* orbitals, while the valence band consists of halide *p* orbitals and some contribution from Pb *s* orbitals. By substituting halides, the energy levels of these orbitals are adjusted, thus tuning the bandgap.^[61] To address the necessity for a comprehensive analysis of halogen substitution strategies, a detailed comparison of their respective advantages and challenges is imperative. A primary advantage of halogen substitution lies in its relatively straightforward approach to bandgap tuning, which allows for the precise tailoring of optical and electronic properties to suit specific applications.^[62] Furthermore, the perovskite crystal structure demonstrates a remarkable ability to maintain its integrity even under significant bandgap modifications, ensuring material stability and adaptability. Another key advantage is the ability of halogen substitution to enhance charge carrier mobility and reduce trap states by optimizing the crystal lattice.^[63]

For instance, bromine substitution often improves the photoluminescence quantum yield, contributing to higher device efficiencies in light-emitting and photovoltaic applications. Chlorine substitution, on the other hand, is particularly effective in enhancing moisture resistance and thermal stability, making it suitable for applications requiring long-term durability under challenging environmental conditions.^[64] Additionally, halogen substitution offers a pathway to color tunability in light-emitting devices, allowing for the design of LEDs with emission across a broad spectrum, including pure green and blue emissions. This versatility is a significant advantage for next-generation displays and lighting technologies.^[65]

Combined with the flexibility to achieve mixed halide compositions, halogen substitution strategies enable the creation of customized material systems for diverse optoelectronic applications, such as tandem solar cells, photodetectors, and lasers.^[66–68] However, notable limitations exist. The permissible alloying ratio of halides within the lattice is inherently constrained, and exceeding this threshold frequently induces phase segregation, adversely affecting material uniformity and performance.^[69] For instance, bromine substitution is often restricted by the emergence of iodide-rich and bromide-rich domains when the halide composition surpasses a critical threshold. This halide segregation compromises lattice uniformity, resulting in suboptimal electronic properties and diminished device efficiency. Similarly, while chlorine substitution has the potential to reduce lattice constants and enhance stability, it encounters significant challenges with phase separation at elevated concentrations, emphasizing the necessity for precise compositional control.^[70] A thorough evaluation of the advantages and limitations associated with each halogen substitution strategy will facilitate the tailored design of perovskite materials, addressing the specific requirements of diverse optoelectronic applications.^[71]

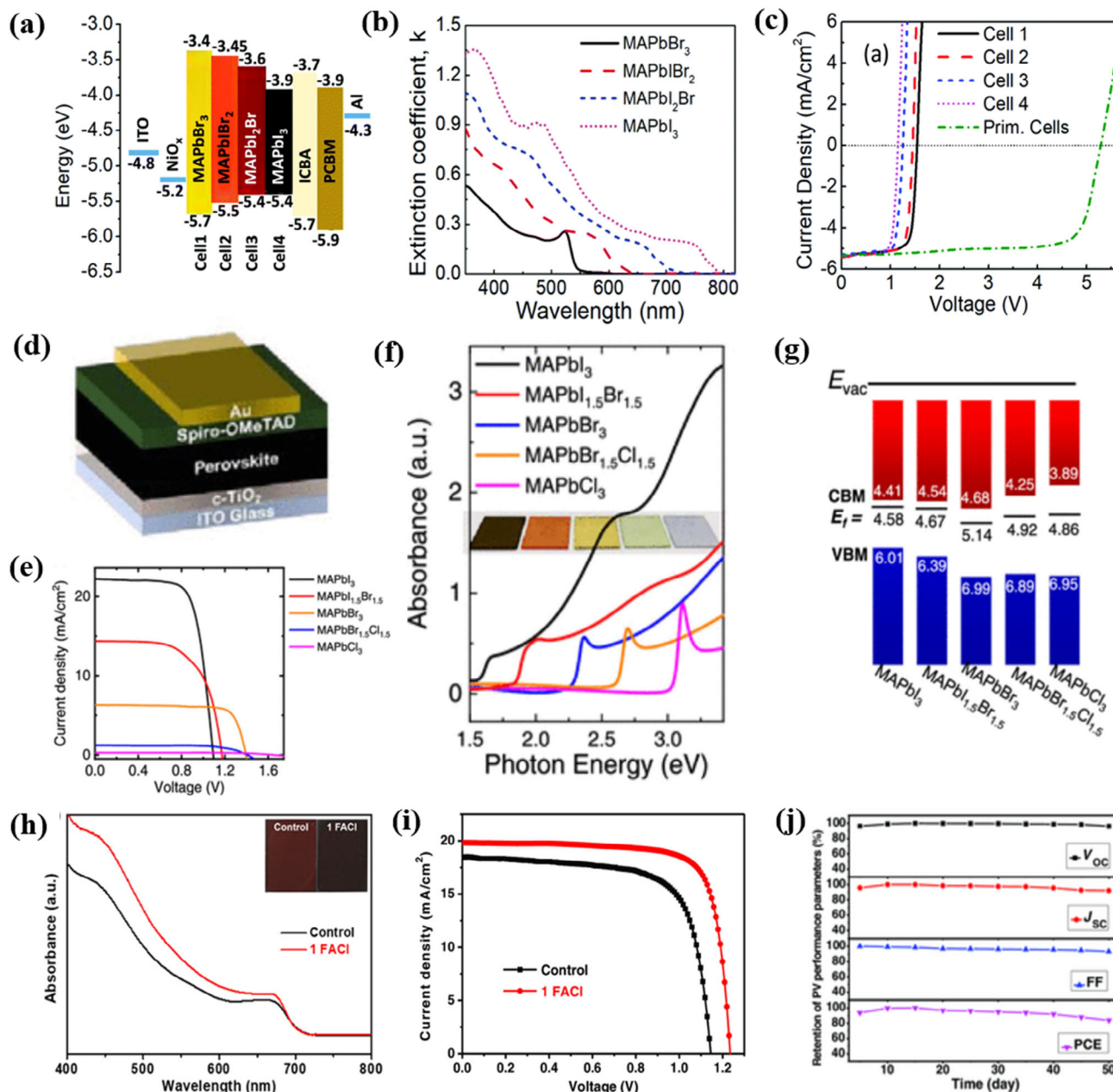


Figure 1. Wide bandgap perovskite solar cell through halide change. a) Energy level diagrams of MAPbI_{3-x}Br_x solar cells. b) extinction coefficients of PVK films including MAPbBr₃, MAPbIBr₂, MAPbI₂Br, and MAPbI₃ and c) J–V characteristics of MAPbI_{3-x}Br_x. a–c) are reprinted with permission from,^[72] copyright 2019 Royal Society of Chemistry. d) Schematic diagram of the wide bandgap solar cell devices. e) J–V curve for the peak efficiency of mixed-halide perovskite solar cells under AM1.5G illumination. f) UV–vis absorption spectra of mixed-halide perovskite films. Inset: photographs of each perovskite thin film. g) Schematics of the electronic structure of the mixed-halide perovskite films. d–g) are reprinted with permission from,^[74] copyright 2021 WILEY-VCH. h) Optical images and UV–vis absorption spectra of FA_{0.83}Cs_{0.17}PbI_{1.8}Br_{1.2}. i) J–V curves of FA_{0.83}Cs_{0.17}PbI_{1.8}Br_{1.2} perovskite solar cell. j) Variation of J–V performance parameters as a function of exposure time of perovskite solar cells based on sample 1 FACI in the ambient (≈30% relative humidity, RT). h–j) are reprinted with permission from,^[75] copyright 2021 WILEY-VCH.

Energy level diagrams of MAPbI_{3-x}Br_x solar cells (Figure 1a) show the shift in energy levels due to halide substitution. As the bromine content increases, the bandgap widens, shifting the CBM upward and the VBM downward, which is essential for increasing the V_{oc} in solar cells. The optical energy bandgaps of

MAPbBr₃, MAPbIBr₂, MAPbI₂Br, and MAPbI₃ films were calculated from their extinction coefficients as 2.3, 2.05, 1.8, and 1.5 eV, respectively. The extinction coefficients of various perovskite films, such as MAPbBr₃, MAPbIBr₂, MAPbI₂Br, and MAPbI₃, are shown in Figure 1b. Higher extinction coefficients

for MAPbBr₃ indicate better light absorption in the blue region, which benefits overall photovoltaic performance. The *J*–*V* characteristics in Figure 1c demonstrate the photovoltaic performance metrics such as Short-circuit current (*J*_{SC}), *V*_{OC}, and fill factor (*FF*) of perovskite solar cells with different halide compositions. The introduction of bromine increases the *V*_{OC} by widening the bandgap, reducing recombination losses, and enhancing charge separation efficiency.^[72]

The device architecture in Figure 1d, adapted from *n-i-p*, includes a TCO glass substrate with a compact titanium dioxide (c-TiO₂) electron transport layer (ETL). The perovskite active layer is followed by a Spiro-OMeTAD hole transport layer (HTL) and a gold (Au) electrode. The current density-voltage (*J*–*V*) characteristics of MAPbI_xBr_yCl_z perovskite solar cells made with these structures provide important insights into performance metrics such as *J*_{SC}, *V*_{OC}, *FF*, and overall power conversion efficiency (PCE). By analyzing the *J*–*V* curves of different halide compositions, it is evident that replacing iodine with bromine and chlorine significantly impacts the photovoltaic parameters, particularly the *V*_{OC} and *J*_{SC}. For MAPbI₃, *V*_{OC} is relatively low (≈1.04 V) but paired with a high *J*_{SC} (≈21.88 mA cm^{−2}), resulting in superior overall efficiency. Conversely, as the halides shift to bromine and chlorine (e.g., MAPbI_{1.5}Br_{1.5}, MAPbBr₃, MAPbBr_{1.5}Cl_{1.5}, and MAPbCl₃), *V*_{OC} increases steadily, reaching as high as 1.65 V in MAPbCl₃. However, this comes at the expense of a dramatic decline in *J*_{SC}, from 13.85 mA cm^{−2} in MAPbI_{1.5}Br_{1.5} to only 0.26 mA cm^{−2} in MAPbCl₃, highlighting a trade-off between voltage and current density with increasing bandgap. The UV–vis absorption spectra shown in Figure 1f highlight the optical properties of mixed-halide perovskite films. As the halide composition transitions from iodine to bromine and subsequently to chlorine, the absorption onset exhibits a blue shift, reflecting an increase in the bandgap energy. The spectra span from 420 to 800 nm, demonstrating strong absorption in the visible range. Notably, the film color evolves from deep black to orange and then to a translucent yellow as the Br or Cl content in the perovskite structure increases, resulting in greater transparency. This gradual substitution of I with Br and then Cl leads to a progressive increase in the bandgap energy from ≈1.6 to 3.1 eV. This reduction in *J*_{SC} is significantly influenced by charge-transport properties. As the halide shifts from iodide to bromide and chloride, carrier mobility decreases due to stronger Fröhlich interactions and increased ionicity of the Pb-halide bond, which enhances carrier-lattice coupling and limits free charge transport. This decline in mobility leads to inefficient charge extraction and increased recombination, further exacerbating the reduction in *J*_{SC}.^[73]

This tunability of the perovskite bandgap through halide substitution is crucial for optimizing light-harvesting efficiency and matching the absorption properties to the solar spectrum. Energy level diagrams in Figure 1g show the VBM and CBM for perovskite materials with different halide compositions: MAPbI₃, MAPbI_{1.5}Br_{1.5}, MAPbBr₃, MAPbBr_{1.5}Cl_{1.5}, and MAPbCl₃. As the halide composition transitions from iodine to bromine, the ionization potential increases from ≈6.0 to 7.0 eV. With chlorine components, the VBM shows little variation, while the CBM shifts to lower values with increasing chlorine concentration. This shift in energy levels affects the charge transport dynamics and the overall efficiency of the solar cells.^[74] To enhance the thermal stability of perovskites by substituting cation with Cs.^[67]

Dong et al. improved FA_{n−8}Cs_{n−2}Pb(I_{n−6}Br_{n−4})₃ perovskite with azetidinium iodide (AZI), achieving 19.16% efficiency.^[75] Building on this, Qiao et al. further enhanced Cs_{n−2}FA_{n−8}Pb(I_{n−6}Br_{n−4})₃ by incorporating potassium hypophosphite (KH₂PO₂), increasing efficiency to 20.06%.^[76]

Despite these advancements, halide substitution in perovskite solar cells faces several limitations. One major challenge is phase segregation, where different halide ions may separate under illumination or thermal stress, forming halide-rich domains. This segregation can create non-uniformities in the material, leading to reduced efficiency and stability. Additionally, while bromine substitution improves thermal stability, it can also increase the volatility of the perovskite films, potentially affecting long-term device performance. To solve these problems, additives are introduced into perovskite solar cells. It is known that the performance and stability of the device are greatly improved when additives are introduced. In particular, the addition of formamidinium chloride (FACl) has proven to be an important advance in the development of next-generation chemicals.

The UV–vis absorption spectra in Figure 1h show the optical properties of the perovskite film with and without FACl additive. The absorption spectrum ranges from 400 to 800 nm, indicating significant absorption in the visible region. Samples with FACl additive exhibit higher absorption intensity and show improved light harvesting capabilities due to improved film quality and better crystal formation. These improvements are important for optimizing the efficiency of perovskite solar cells by maximizing light absorption. The *J*–*V* characteristics in Figure 1i reveal that perovskite solar cells with the FACl additive achieve higher *J*_{SC} of ≈20 mA cm^{−2} and *V*_{OC} of ≈1.1 V, compared to the control device. These enhancements are due to better crystallinity and reduced defect densities in the FACl-treated films, leading to more efficient charge separation and transport. Figure 1j shows the stability of these cells over 50 days. The FACl-enhanced device retains ≈83.8% of its original PCE after this period, indicating superior long-term stability compared to the control. This increased stability results from reduced ion migration and improved film quality achieved through intermediate phase engineering with FACl.^[77]

In a similar vein, Snaith et al. demonstrate the use of chloride-based additive engineering to improve the efficiency and stability of wide-bandgap perovskite solar cells. By incorporating methylammonium chloride (MACl) into the perovskite crystallization process, they achieved a maximum PCE of 17.0% and a *V*_{OC} of 1.25 V, and a bandgap of 1.63 eV highlighting the role of chloride in enhancing halide homogenization and improving the overall optoelectronic properties of the material.^[78] Further extending these advancements, Kanatzidis et al. demonstrated that incorporating dimethylammonium (DMA) and acetamidinium (Ac) into mixed-halide perovskites led to significant improvements in the material's properties. They achieved a bandgap of 1.73 eV and a PCE of 16.2%.^[79] This approach improved the photostability and overall performance of the perovskite solar cells, making them more suitable for tandem cell applications. Additionally, Yi et al. developed a self-assembled monolayer of (4-(5,9-dibromo-7H-dibenzo[*c,g*]carbazol-7-yl)butyl)phosphonic acid (DCB-BPA) as a hole-selective layer for WBG FA_{n−8}Cs_{n−2}PbI_{1−8}Br_{1−2} perovskite solar cells. This innovation enhanced the buried interface quality, leading to a high *V*_{OC} of 1.339 V and a PCE of 18.88%. The device demonstrated low *V*_{OC} loss of 431 mV, improved

photostability, and achieved a 4-terminal tandem cell with a PCE of 26.9%, showcasing its potential in perovskite tandem applications.^[80]

Halide substitution in perovskite solar cells, especially replacing iodine with bromine or chlorine, significantly improves the electronic structure and stability of the material by increasing the band gap and improving thermal stability. Energy level diagrams and UV–vis absorption spectra confirm a wider bandgap and better light absorption in the visible range, which is important for optimizing the efficiency of these solar cells. However, challenges remain, such as phase separation and increased film volatility. To solve these problems, effective control methods using various additives are being actively researched. These developments highlight the important role of halide compositions and additives in the development of highly efficient, stable perovskite solar cells.

3. Wide Bandgap Perovskite Through Inorganic Cation

The substitution of the A-site cation with cesium (Cs^+) in lead triiodide perovskites significantly increases the bandgap due to its smaller ionic radius, which leads to a more compact lattice structure and greater orbital overlap between Pb and I atoms. This compaction stabilizes the CBM and increases the bandgap, enhancing the V_{OC} of solar cells.^[81] Theoretical studies and experimental data confirm that CsPbI_3 exhibits higher CBM energy levels and a larger bandgap compared to MAPbI_3 and FAPbI_3 , consistent with observed temperature and pressure dependencies.^[82] This increase in bandgap, while improving V_{OC} and thermal stability, may reduce the absorption of lower-energy photons, impacting J_{SC} .^[83] Overall, the use of Cs^+ as the A-site cation optimizes the electronic properties and enhances the performance and stability of perovskite solar cells.^[84]

Chong et al. introduce dimethylammonium iodide (DMAI) into the co-evaporation process to produce $\text{DMA}_{1-x}\text{Cs}_x\text{PbI}_3$ films with reduced defects and enhanced stability, achieving efficiencies comparable to high-temperature processed CsPbI_3 films. Figure 2a is X-ray diffraction patterns indicate that DMAI incorporation leads to a preferential crystallographic orientation and larger grain size, reducing defects and improving overall film quality. The photoluminescence spectrum (Figure 2b) shows a decrease in the bandgap from 1.76 eV for $\gamma\text{-CsPbI}_3$ to 1.73 eV for $\text{DMA}_{0.06}\text{Cs}_{0.94}\text{PbI}_3$. This is due to less lattice distortion and increased Pb–I overlap by replacing Cs^+ with DMA^+ . This red shift enhances the spin-orbit coupling effect and narrows the bandgap. J – V curves (Figure 2c) show that the $\text{DMA}_{0.06}\text{Cs}_{0.94}\text{PbI}_3$ -based device achieves higher J_{SC} of $\approx 20 \text{ mA/cm}^2$ and V_{OC} of $\approx 1.1 \text{ V}$ compared to the control, resulting in a record efficiency of 16.10%. These improvements are attributed to better crystallinity and fewer defects. Stability tests indicate that $\text{DMA}_{0.06}\text{Cs}_{0.94}\text{PbI}_3$ devices retain $\approx 83.8\%$ of their original PCE after 50 days, thanks to reduced ion migration and improved film quality, which enhance structural stability, humidity resistance, and phase stability.^[85]

Xu et al. studies investigate how adding ethylammonium (EA) to cesium lead triiodide (CsPbI_3) perovskite solar cells affects their bandgap, stability, and charge transport properties. The device architecture depicted in Figure 2d is a p–i–n structure, consisting of a glass substrate with fluorine-doped tin oxide (FTO), a compact TiO_2 as the ETL, a perovskite active layer,

and a spiro-OMeTAD as the HTL, and silver (Ag) electrodes. Introducing EA into the perovskite structure alters the crystallization process, leading to improved phase stability and better transport properties. The tolerance factor, a key measure for phase stability calculated as $t = (r_A + r_X)/\sqrt{2(r_B + r_X)}$, where r_A , r_X , and r_B are the ionic radii of the A-site cation, B-site metal, and halide ion respectively, increases with the addition of EA.^[86] Pure CsPbI_3 has a low tolerance factor of 0.81, making it less stable at room temperature, but adding EA, which has a larger ionic radius, enhances the stability of the perovskite structure, as shown in Figure 2e. Photoluminescence (PL) spectra and Tauc plots in Figure 2f reveal that incorporating EA into CsPbI_3 slightly increases the bandgap from 1.68 eV to over 1.7 eV. This shift is due to the distortion of the Pb–I–Pb bond angle, which changes the electronic structure by increasing the antibonding interaction between Pb 6p and I 5p orbitals. The J – V characteristics shown in Figure 2g highlight the performance metrics of perovskite solar cells with different EA fractions. Notably, devices with $\text{EA}_{0.1}\text{Cs}_{0.9}\text{PbI}_3$ exhibit higher J_{SC} and V_{OC} compared to pure CsPbI_3 . Specifically, the $\text{EA}_{0.1}\text{Cs}_{0.9}\text{PbI}_3$ -based device achieves a J_{SC} of $\approx 20 \text{ mA cm}^{-2}$ and a V_{OC} of $\approx 1.1 \text{ V}$, resulting in a PCE of 16.01%, attributed to better crystallinity and fewer defects. Additionally, perovskite solar cells using Cs^+ as the A-site cation can further adjust the bandgap through halide exchange.^[87]

Incorporating Br and I into perovskite compositions plays a crucial role in determining their bandgap and efficiency. For example, Zhou et al. demonstrated that CsPbI_2Br , with a bandgap of $\approx 1.89 \text{ eV}$, achieves a PCE of $\approx 17.33\%$.^[88] This composition offers enhanced thermal stability and reduced recombination losses, making it well-suited for high-temperature applications. Conversely, Hu et al. reported that CsPbIBr_2 , which has a slightly wider bandgap of 2.08 eV, achieves a higher PCE of 9.38%, as it effectively balances light absorption and charge transport properties.^[89] By carefully optimizing the halide composition, researchers can adjust the electronic properties of perovskite solar cells, thereby enhancing their overall performance and stability. Moreover, Yang et al. introduced a chlorosilane molecule (1,2-bis(chlorodimethylsilyl)ethane, CSE) into the perovskite precursor to enhance the performance of all-inorganic CsPbI_3 perovskite solar cells. This approach effectively passivated Pb-related defects, suppressed non-radiative recombination, and optimized the energy level alignment. The resulting device achieved a PCE of 21.8%, showcasing one of the highest efficiencies reported for CsPbI_3 perovskites.^[90]

To expand the discussion on WBG PSCs and demonstrate the possibility of substituting organic cations or elements other than the cation Cs in A-site, researchers investigated various compositions and their effects on bandgap and efficiency. For instance, rubidium (Rb) has been investigated for its potential to replace Cs in perovskite structures. Moon et al. investigate the improvements in perovskite solar cells by using rubidium lead iodide (RbPbI_3) to enhance stability and performance under environmental conditions. The energy level diagram in Figure 2h illustrates that the band alignment of RbPbI_3 with TiO_2 is well-suited for efficient charge transfer, with compatible energy levels throughout the device layers. The fluorescence spectra in Figure 2i reveal that RbPbI_3 exhibits significant exciton binding energy due to its quasi-2D PbI_6 octahedral network, resulting in strong

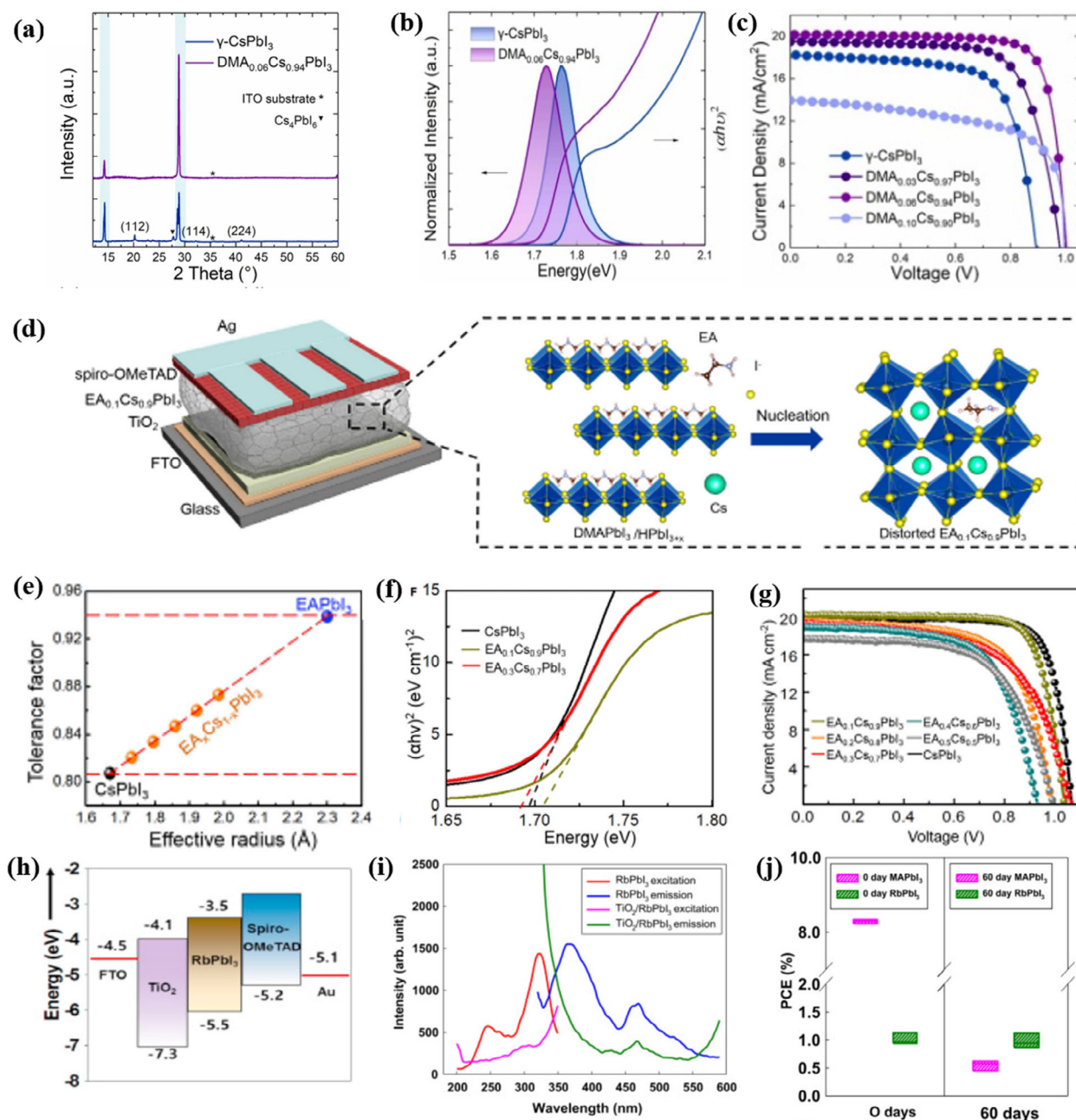


Figure 2. Wide bandgap perovskite solar cell through inorganic cation. a) XRD patterns for γ -CsPbI₃ and DMA_{1-x}Cs_xPbI₃ thin films b) The steady-state PL spectra and the UV-vis absorption spectra for the perovskite films with or without DMAI. c) The J-V curves for the γ -CsPbI₃ and DMA_{1-x}Cs_xPbI₃ perovskite solar cells. a–c) are reprinted with permission from,^[85] copyright 2024 Science Direct. d) The schematic of the EA_xCs_{1-x}PbI₃ perovskite solar cell. e) Tolerance factors of the EA_xCs_{1-x}PbI₃ perovskite films. f) Tauc plots of the EA_xCs_{1-x}PbI₃ perovskite films. g) J-V curve of EA_xCs_{1-x}PbI₃ perovskite solar cells. d–g) are reprinted with permission from,^[87] copyright 2021 Science Direct. h) Energy level for the FTO/TiO₂/RbPbI₃/spiro-MeOTAD/Au solar cell configuration. i) Fluorescence spectra of RbPbI₃ and RbPbI₃/TiO₂ films j) The performance comparison between 10 MAPbI₃ cells and 10 RbPbI₃ cells in terms of device stability was tested before and after being exposed to a relative humidity of 40% at 25 °C for 60 days. h–j) are reprinted with permission from,^[88] copyright 2021 Science Direct.

photoluminescence and a bandgap of ≈ 1.9 eV, influenced by the antibonding interaction between Pb 6p and I 5p orbitals. Emission peaks affected by the TiO₂ layer indicate effective charge transfer between TiO₂ and RbPbI₃ upon light exposure. Figure 2j) presents a comparison of the stability per-

formance between 10 MAPbI₃ cells and 10 RbPbI₃ cells, tested before and after 60 days of exposure to 40% relative humidity at 25 °C. The results show that while the performance of MAPbI₃ cells significantly deteriorates, the RbPbI₃ cells maintain $\approx 83.8\%$ of their initial PCE, demonstrating superior

stability due to their robust crystal structure and reduced ion migration.^[91]

Inorganic cations, like Cs⁺ and Rb⁺, are key to enhancing the performance and stability of perovskite solar cells. Cs⁺ increases the bandgap and thermal stability by creating a more compact lattice structure, while Rb⁺ offers superior environmental stability and reduced ion migration. Additionally, additives such as DMAI and EA improve film quality by reducing defects and enhancing crystallinity. These strategies collectively lead to better charge transport, higher power conversion efficiencies, and greater long-term stability, making them essential for advancing perovskite solar cell.

4. Wide Bandgap 2D/3D Perovskite

The shift from 3D to 2D perovskite structures brings notable enhancements in material properties, particularly concerning stability and bandgap tuning.^[92–94] While 3D perovskites consist of a network of corner-sharing octahedra that provide high carrier mobility and robust light absorption, they often struggle with long-term stability issues due to susceptibility to moisture and thermal degradation.^[95] In contrast, 2D perovskites, characterized by the formula (RNH₃)₂A_{n-1}B_nX_{3n+1}, benefit from large organic spacer cations that create protective barriers, enhancing environmental stability.^[96,97] The widening of the bandgap in 2D perovskites compared to 3D counterparts is mainly due to quantum and dielectric confinement effects.^[98] Reduced dimensionality in 2D perovskites confines charge carriers within the thin inorganic layers, increasing the energy difference between the CBM and VBM.^[99] Additionally, the lower dielectric constant of the organic layers intensifies this confinement, leading to a larger bandgap, as evidenced by the significant blue shift in photoluminescence (PL) emissions with thinner inorganic layers.^[100]

However, the introduction of 2D layers can impede charge transport across the organic-inorganic interfaces, potentially lowering the overall photovoltaic performance compared to pure 3D perovskites.^[101] To address these issues, mixed-dimensional 2D/3D perovskite structures have been developed.^[102–105] These hybrids combine the stability of 2D layers with the superior electronic properties of 3D WBG perovskites, achieving high PCE above 21% and maintaining stability under prolonged exposure to ambient conditions.^[106] Despite these advancements, optimizing 2D/3D perovskite solar cells requires precise control over synthesis and processing conditions to balance stability and performance.^[107] Further research into the fundamental mechanisms of phase transitions and defect passivation in these materials is essential. Building on this, Xia et al. have developed quasi-2D PSCs enhanced with a fluorinated additive, 3-(trifluoromethyl) benzylammonium iodide (3-TFMBAI), to improve environmental stability and photoelectric performance. The additive promotes the formation of the Ruddlesden-Popper phase, enhancing crystallinity, passivating trap states, and suppressing degradation due to humidity and heat, resulting in a peak efficiency of 20.89% and long-term stability under challenging conditions. Four-terminal tandem solar cells combining these quasi-2D PSCs with silicon solar cells achieved an impressive efficiency of 23.53%.^[108] In addition, Huang et al. demonstrated a 2D/3D perovskite heterostructure stacking strategy to mitigate interfacial defects and suppress non-radiative

recombination in wide-bandgap (WBG) perovskite solar cells, achieving a record efficiency of 23.05%. By employing a dual-ammonium approach, the study optimized surface lattice structures and achieved notable efficiencies of 21.25% for large-area cells and 20.10% for mini-modules, highlighting its potential for commercialization.^[109]

In this context, Song et al. employed octadecylamine (ODA) as Target 1 and butylammonium iodide (BAI) as Target 2 to enhance 2D/3D perovskite solar cells. The devices treated with BAI exhibited superior efficiency, which can be attributed to the ability of BAI to effectively passivate surface defects and form a stable interface, leading to improved overall device performance. **Figure 3a** shows the electroluminescence (EL) spectra of the device treated with Target 2 across various applied voltages (1.5–2.0 V). The increased EL intensity with higher voltage indicates efficient charge injection and radiative recombination. This enhancement is due to the high-quality film formation and reduced non-radiative recombination pathways provided by the BAI treatment, resulting in better light emission properties. The *J*–*V* curves in **Figure 3b** compare the performance of control, Target 1, and Target 2 devices. The Target 2 device achieves a PCE of 22.19%, outperforming the control (18.50%) and Target 1 (21.36%). This significant improvement is due to the superior crystallinity and charge transport properties facilitated by the BAI treatment, which reduces charge recombination losses and enhances overall photovoltaic performance. **Figure 3c** presents the stability test results, showing the normalized PCE of the devices over time at 60°C. The Target 2 device maintains 95% of its initial PCE after 1000 h, demonstrating excellent long-term stability. In contrast, the control device's efficiency drops to zero within the same period. This stark difference highlights the enhanced thermal stability and durability imparted by the BAI treatment, which effectively prevents ion migration and phase segregation, preserving the structural and electronic integrity of the perovskite film under prolonged thermal stress.^[44] Cahen et al. utilized hexyltrimethylammonium bromide (HTAB) as an interfacial modifier to enhance the performance and stability of carbon-based CsPbIBr₂ perovskite solar cells. The choice of HTAB is strategic, leveraging its capability to passivate surface defects and improve the electronic properties of the perovskite layer. The trimethylammonium group (N⁺(CH₃)₃) stabilizes the perovskite structure by passivating uncoordinated Pb²⁺ sites, while the hydrophobic alkyl chain (C₆H₁₃) enhances moisture resistance, contributing to overall device stability.

Figure 3d shows the device architecture without HTL, which includes a glass substrate with fluorine-doped tin oxide (FTO), a small TiO₂ ETL, a CsPbIBr₂ perovskite layer, an HTAB interfacial layer, and a carbon electrode. The HTAB treatment results in a more uniform and defect-free perovskite film. The contact angle measurements in **Figure 3e** show a significant increase in the contact angle for the HTAB-treated film (69.6°) compared to the pristine film (33.1°), indicating enhanced hydrophobicity and improved moisture resistance. **Figure 3f** displays the photoluminescence (PL) spectra and absorption data, demonstrating that the HTAB-treated films have stronger PL intensity and slightly higher absorption in the visible range. This indicates reduced non-radiative recombination and better crystallinity due to the HTAB treatment. The *J*–*V* curves in **Figure 3g** reveal that the HTAB-treated device achieves a higher PCE of 11.24%, outperforming the pristine device. This improvement is attributed to

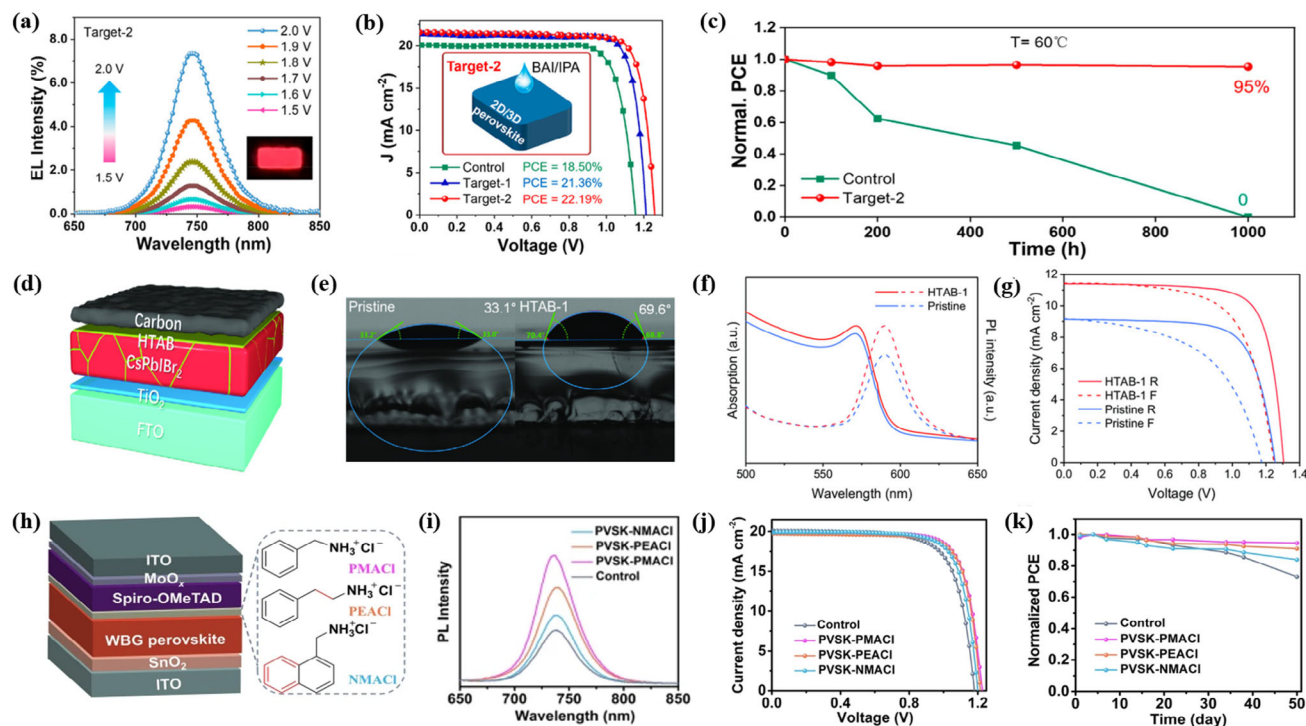


Figure 3. Wide bandgap 2D/3D perovskite solar cell. a) EL spectra of the target-2 device ($\text{BAI-FA}_{0.83}\text{Cs}_{0.17}\text{PbI}_{1.8}\text{Br}_{1.2}$) operated as a light-emitting diode (LED) under different voltage biases. b) J - V curves of the control, target-1, and target-2 devices. b) J - V curves of the control, target-1 ($\text{ODAI-FA}_{0.83}\text{Cs}_{0.17}\text{PbI}_{1.8}\text{Br}_{1.2}$), and target-2 devices. c) Comparison of the stability of normalized PCE for the control and target-2 devices stored at 60°C in N_2 . a–c) are reprinted with permission from,^[44] copyright 2023 American Chemical Society. d) Schematic device structure and i) contact angles of water droplets on the pristine and HTAB-1 perovskite films. f) Absorption and steady-state PL spectra and g) J - V curves of the pristine and HTAB-1 PSCs under reverse and forward scanning directions. d–g) are reprinted with permission from,^[110] copyright 2023 WILEY-VCH. d) Schematic of the device and the chemical structure of PMACl, PEACl, and NMACl. i) The steady-state PL spectra. j) J - V curves the WBG-PSCs and k) stability of the WBG-PSCs stored at $\approx 25^\circ\text{C}$ in air with a relative humidity of $\approx 10\%$. h–k) are reprinted with permission from,^[111] copyright 2023 Springer Link.

the enhanced film quality, reduced defect density, and improved charge transport facilitated by the HTAB modification.^[110]

Liu et al. investigated the use of three bulky cation chlorides—phenylmethylammonium chloride (PMACl), phenylethylammonium chloride (PEACl), and *n*-methylammonium chloride (NMACl) to enhance the surface passivation of wide-bandgap (WBG) perovskite solar cells. The primary goal was to reduce non-radiative recombination and address the V_{OC} deficit. Each chloride was chosen for its potential to form effective passivation layers with the perovskite film. The use of these chlorides forms distinct passivation layers, each contributing differently to the device's performance. PMACl treatment results in a pure chloride 2D perovskite layer, $(\text{PMA})_2\text{PbCl}_4$, known for its excellent passivation properties due to high purity and structural dimensionality. PEACl forms a mixed-anion 2D perovskite layer, $(\text{PEA})_2\text{PbI}_{x-1}\text{Cl}_{4-x}$, which also provides good surface passivation but is slightly less effective than PMACl. NMACl forms a non-2D NMAI salt passivation layer, which helps in reducing surface defects but does not perform as well as PMACl and PEACl.

The device architecture, illustrated in Figure 3h, includes a stack of ITO as the bottom electrode, SnO_2 as the ETL, the WBG perovskite layer, the passivation layer (treated with PMACl, PEACl, or NMACl), Spiro-OMeTAD as the HTL, MoO_x , and an additional ITO layer as the top electrode. The photoluminescence (PL) spectra in Figure 3i show that the PL intensity significantly

increases for the films treated with PMACl and PEACl, indicating a reduction in non-radiative recombination. The PMACl passivated perovskite film shows the highest PL intensity, suggesting the most effective passivation. The J - V curves in Figure 3j demonstrate that the PMACl-treated devices achieve the highest PCE of 18.60% and a V_{OC} of 1.23 V, outperforming the control and other treatments. This enhancement is attributed to the pure 2D $(\text{PMA})_2\text{PbCl}_4$ layer, which serves as an effective electron-blocking barrier, thereby reducing recombination. The stability data in Figure 3k show that PMACl-treated devices retain 95% of their initial PCE after 1000 h at 60°C , highlighting superior long-term stability. This durability is due to the robust passivation provided by the $(\text{PMA})_2\text{PbCl}_4$ layer, which mitigates defect states and improves overall stability. In summary, among the three chlorides tested, PMACl offers the most effective passivation, resulting in higher efficiency and enhanced stability in WBG perovskite solar cells, making it the preferred choice.^[111]

5. Lead-Free Wide Bandgap Perovskite

Currently, PSCs are being touted as the next generation of solar cell technology due to their high efficiency and low production costs.^[112] However, Pb, a key component of these solar cells, poses serious environmental and health concerns.^[113] Lead is highly toxic, causing environmental pollution when solar cells

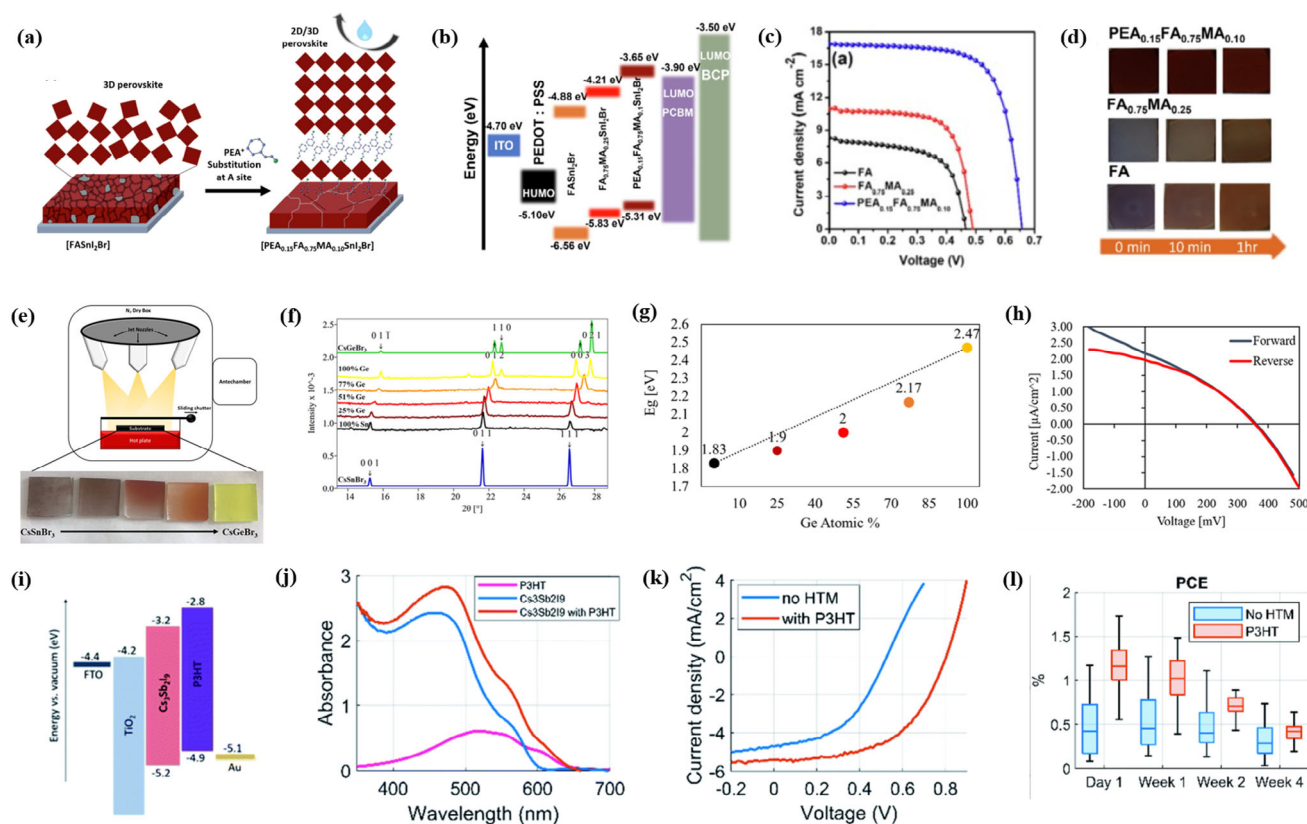


Figure 4. Lead-free wide-bandgap perovskite solar cell. a) Schematic the substitution of PEA⁺ cation at the A site in a mixed cation 3D perovskite and b) energy band alignment for FA, FA_{0.75}MA_{0.25}, and PEA_{0.15}FA_{0.75}MA_{0.10} perovskite device. c) J–V curves of PSCs and d) Degradation and related changes of different perovskite compositions over varying exposure intervals. a–d) are reprinted with permission from,^[125] copyright 2022 American Chemical Society. Science Direct. e) Deposition process and color change of CsSn_xGe_{1-x}Br₃ films. f) XRD patterns for spray-deposited CsSn_xGe_{1-x}Br₃ (x = 0–1) thin films. g) Optical band gaps of CsSn_xGe_{1-x}Br₃ (x = 0–1), derived from the data in the panel. h) J–V curves for CsSn_xGe_{1-x}Br₃ (Sn = 49%, Ge = 51%) device. e–h) are reprinted with permission from,^[126] copyright 2022 American Chemical Society. i) Energy levels diagram for Cs₃Sb₂I₉ perovskite solar cells. j) UV–vis absorbance of the solar cells with and without P3HT. k) J–V curves for solar cells with and without P3HT and l) Statistical analysis of the solar cell stability over one month of storage in air, in the dark, at 20–21°C. i–l) are reprinted with permission from,^[127] copyright 2021 Royal Society of Chemistry.

are disposed of, and lead leaching into soil and water can harm ecosystems and human health.^[114] Therefore, a transition to lead-free perovskite materials is needed for environmentally safe and sustainable energy solutions. Recent research is exploring a variety of lead-free perovskite materials that could replace lead.^[115]

However, lead-free WBG perovskites present several challenges. First, while lead-based perovskites exhibit high efficiency due to their high light absorption and charge transport properties, lead-free perovskites can exhibit relatively low efficiency.^[116] For example, tin (Sn)-based perovskites have narrow bandgap and are more prone to charge recombination compared to lead-based materials.^[117,118] Second, lead-free perovskites are more sensitive to environmental conditions (e.g., humidity, heat), making them less stable. In particular, tin-based perovskites are highly susceptible to oxidation and can quickly decompose in air.^[119] Third, the manufacturing process for lead-free perovskites is complex and requires high-purity materials, which can make it difficult to mass produce and reduce costs.^[120] To address these issues, researchers are exploring various strategies such as substituting different metal cations, halide exchange, and ad-

vancements in processing techniques.^[121–124] By optimizing the composition and processing methods, the stability and efficiency of lead-free perovskites can be significantly improved, paving the way for their practical application in photovoltaic devices.

Kang et al. explored the integration of phenylethylammonium (PEA⁺) cations into 3D perovskites to enhance the performance and stability of PSCs.^[125] The inclusion of PEA⁺ forms a quasi-2D/3D hybrid structure, effectively passivating grain boundaries and reducing moisture ingress, which helps mitigate the oxidation of Sn²⁺ ions. In Figure 4a, the schematic illustrates the substitution of PEA⁺ at the A-site in mixed cation perovskites, resulting in improved crystallinity and preferential orientation. Figure 4b shows the energy band alignment for FA, FA_{0.75}MA_{0.25}, and PEA_{0.15}FA_{0.75}MA_{0.10} perovskite (lead-free) devices, indicating that PEA⁺ incorporation leads to better band alignment, crucial for efficient charge transport. The J–V curves in Figure 4c demonstrate that the device with PEA_{0.15}FA_{0.75}MA_{0.10} achieves a higher PCE of 7.96%, with *J*_{SC} of 16.89 mA cm^{−2} and *V*_{OC} of 0.67 V, outperforming other compositions. Stability tests in Figure 4d highlight that the PEA⁺ incorporated devices exhibit significant

resistance to moisture-induced degradation, maintaining structural integrity and performance over time, unlike FA and FA_{0.75}MA_{0.25} which degrade rapidly. These results emphasize the role of PEA⁺ in forming a protective layer that reduces non-radiative recombination, leading to enhanced efficiency and longevity of lead-free perovskite solar cells.

Kama et al. developed a new lead-free solid solution halide perovskite (CsSn_xGe_{1-x}Br₃) by substituting toxic Pb ions with Sn and Ge.^[126] To overcome the low solubility of low-concentration precursor solutions and improve film coverage, they used a multi-head spray deposition system, forming uniform solid solution CsSn_xGe_{1-x}Br₃ films. This method allowed maintaining a single crystal structure during the composition change from pure Sn to pure Ge. The tetragonal structure of pure CsSnBr₃ was maintained up to 77% Ge content, and as the Ge content increased, the lattice spacing decreased. The optical bandgap was tunable between 1.83 and 2.47 eV from pure Sn to pure Ge. Notably, higher Ge content improved the structural stability of the perovskite by reducing the oxidation of both Ge and Sn in their +4 states. Electrical and electronic transport measurements indicated that these materials are suitable for use in solar cells as Pb-free absorbers, especially given their bandgap range which makes them suitable as the top cell in tandem photovoltaic devices like Si.

In Figure 4e, the deposition process of CsSn_xGe_{1-x}Br₃ films is shown. Using a multi-head spray deposition method, low-concentration precursor solutions form continuous films with better coverage than traditional spin coating methods. This process leads to a visual color change from yellow (pure CsGeBr₃) to brown (pure CsSnBr₃), demonstrating the uniform deposition and compositional variation achieved through this method. Figure 4f presents the X-ray diffraction (XRD) patterns for spray-deposited CsSn_xGe_{1-x}Br₃ (x = 0–1) thin films. As the Ge content increases, the lattice spacing decreases, which causes a shift in the 2θ angle to higher values. For instance, the 2θ peak shifts from ≈21.5° for pure CsSnBr₃ to ≈23.5° for CsSn_{0.25}Ge_{0.75}Br₃, indicating a contraction in the crystal lattice due to the substitution of larger Sn atoms with smaller Ge atoms. Figure 4g illustrates the optical bandgaps of CsSn_xGe_{1-x}Br₃ as a function of Ge content. The bandgap widens from 1.83 eV for pure CsSnBr₃ to 2.47 eV for pure CsGeBr₃. This tunability in the bandgap is crucial for optimizing the material for specific photovoltaic applications, allowing for the design of perovskites with tailored electronic properties. Figure 4h shows the J–V curves for a device with a composition of 49% Sn and 51% Ge. The device exhibits a V_{OC} of 360 mV and a J_{SC} of 2 μA cm⁻², with a PCE of ≈0.1%. These relatively low values suggest issues with film coverage and contact resistance, highlighting areas for further optimization in the deposition process and device fabrication.

Antimony (Sb), with its electronic properties akin to lead, is a promising substitute in perovskite structures. The trivalent nature of Sb forms Cs₃Sb₂I₉, a perovskite derivative with two polymorphic forms: a 0D dimer phase and a 2D layered structure. The 2D form is particularly advantageous for solar applications due to its superior carrier mobility and defect tolerance. Early research on Cs₃Sb₂I₉ perovskites showed promising results for solar cell applications. Initial work by Boopathi et al. achieved a PCE of 0.84% using Cs₃Sb₂I₉-based solar cells, which later improved to 1.5% with a vapor-assisted deposition process.^[127] These advancements were largely due to improved film quality and en-

hanced charge transport. Building on these foundational studies, more recent research has focused on optimizing device architecture and materials to further enhance performance and stability.

The device architecture (Figure 4i) consists of a layered structure starting with an ITO substrate, followed by a TiO₂ ETL, Cs₃Sb₂I₉ perovskite absorber, P3HT HTM, and a gold (Au) electrode. This architecture ensures efficient charge separation and transport, reducing recombination losses. The UV–vis absorbance spectra (Figure 4j) show that the inclusion of P3HT in the solar cells enhances light absorption across the visible spectrum, particularly in the 400–700 nm range, indicating better photon capture and conversion efficiency. The J–V curves (Figure 4k) reveal that the solar cells with P3HT achieve significantly higher performance, with the champion device reaching a PCE of 2.5%. In comparison, devices without P3HT show much lower efficiencies. Recently, a similar method using P3HT as a hole transport material was used to achieve high efficiency (PCE 22.7%) in a lead-based perovskite solar cell. The device was able to maintain 85% of its initial PCE for over 1000 h without encapsulation. The V_{OC}, J_{SC}, and FF improvements are attributed to the efficient hole transport and reduced energy loss at the perovskite/HTM interface provided by P3HT. The stability of the devices (Figure 4l) was monitored for one month under ambient conditions. During this period, devices with P3HT showed a 66% decrease in PCE. This decrease was most pronounced in the J_{SC} and V_{OC}, indicating initial instability of P3HT. While P3HT can be stable when combined with suitable counterparts, it appears that Cs₃Sb₂I₉ does not adequately protect P3HT. P3HT undergoes rapid photodegradation, but this can be mitigated when mixed with the right acceptor materials. There is also a possibility of unwanted reactions between Cs₃Sb₂I₉ and P3HT. Addressing these issues may require modifying the P3HT-perovskite interface. Using Sb in the form of Cs₃Sb₂I₉ leverages Sb's ability to form stable, lead-free perovskite structures with suitable bandgaps for solar energy conversion. The octahedral coordination of Sb with halides I leads to the formation of a perovskite structure that can efficiently absorb sunlight and generate charge carriers. Additionally, Sb-based perovskites exhibit lower toxicity and better environmental compatibility compared to their Pb-based counterparts.

However, Sb-based perovskites also face several limitations. One significant challenge is their relatively low power conversion efficiency compared to Pb-based perovskites, partly due to less optimal charge transport properties.^[128] Moreover, the stability, while improved compared to Pb-based perovskites, still requires further enhancement to ensure long-term durability under operational conditions.^[129] Environmental stability, particularly against moisture and oxygen, remains a critical concern. There is also a need for better understanding and optimization of the interfaces between Sb-based perovskites and other layers in the solar cell to minimize recombination losses and improve charge extraction.

6. Wide Bandgap Perovskite-Like Materials

Rudorffite materials, most notably AgBiI₄, have gained substantial attention as promising lead-free alternatives for stable solar cells.^[130] The pursuit of these materials is largely driven by the need to mitigate the environmental and health risks associated

with lead-based perovskites. Rudorffites are defined by their general formula $A_xB_yX_{x+3y}$, where A and B are metal cations, and X is a halide anion.^[131] In the case of $AgBiI_4$, the structure comprises silver (Ag) as the monovalent cation and bismuth (Bi) as the trivalent cation, with iodide (I) serving as the halide. This configuration forms a cubic-close-packed iodide sublattice with Ag and Bi occupying edge-sharing octahedral sites, creating a 3D network where all tetrahedral sites remain unoccupied. This cubic defect spinel structure, with its specific space group and lattice parameters, provides a solid and stable framework. The 3D network of edge-sharing $[AgI_6]$ and $[BiI_6]$ octahedra significantly enhances the material's thermal and photostability. Integrating silver cations into the iodide sublattice increases the dimensionality of iodobismuthates, thereby improving their semiconducting properties.^[132,133]

Bismuth plays a critical role in replacing lead due to its similar electronic configuration and significantly lower toxicity.^[134] Bismuth-based compounds are known for their stable structures and relatively smaller band gaps, which make them suitable for photovoltaic applications. In $AgBiI_4$, the band gap is ≈ 1.86 eV, which is closer to the ideal range for effective light absorption in solar cells compared to many other Bi-based compounds.^[135,136] Bismuth also contributes to the material's structural stability, unlike Sn and Ge, which are prone to oxidation and result in poor stability.^[137] The consistent oxidation state of bismuth enhances the material's durability under operational conditions, which is crucial for the long-term performance of solar cells.^[138]

The role of orbitals in $AgBiI_4$ is pivotal to understanding its electronic properties. Both bismuth and silver, being heavy elements, exhibit significant spin-orbit coupling, influencing their electronic structure. The VBM primarily consists of Bi 6p and I 5p orbitals, while the CBM includes Ag 5s and Bi 6p orbitals. The strong interaction between Bi 6p and I 5p orbitals lowers the band gap, making $AgBiI_4$ an effective light absorber.^[131] These properties highlight $AgBiI_4$'s potential as a promising material for solar energy harvesting. However, despite these advantages, Rudorffite materials like $AgBiI_4$ face notable efficiency challenges due to their inherent *p*-type conductivity. This limits the mobility of excited electrons, leading to slower charge transport and higher non-radiative recombination rates.^[139] Furthermore, high trap-state densities and structural defects exacerbate these issues, while their relatively low light absorption coefficients make it difficult to achieve high photovoltaic performance compared to lead-based perovskites.^[140] Suggesting that further optimization is needed to fully capitalize on their potential for photovoltaic applications.

The flexibility of the rudorffite structure also allows for substitution with other elements to further tailor its properties. For instance, the monovalent cation Ag can be replaced by other monovalent cations like cesium (Cs), Rubidium (Rb), or potassium (K), while bismuth can be substituted with other trivalent cations such as antimony (Sb).^[141,129] These substitutions can lead to new materials with different band gaps, stabilities, and optoelectronic properties. This versatility makes the rudorffite family a rich field for discovering new materials with optimized properties for specific applications.

Another promising class of materials in this context is the double perovskites, which have the general formula $A_2B'B''X_6$, where A is a monovalent cation, B' and B'' are metal cations, and X is

a halide anion.^[142] Double perovskites, such as $Cs_2AgBiBr_6$ and $Cs_2AgBiCl_6$, have been explored as lead-free alternatives due to their inherent stability and suitable band gaps for photovoltaic applications.^[143] These materials exhibit different electronic and structural properties compared to traditional perovskites and rudorffites, offering a diverse range of characteristics that can be fine-tuned for enhanced performance in solar cells.

Future research on optimizing $AgBiI_4$ -based solar cells will focus on enhancing PCE and understanding the mechanisms of charge transport and recombination in detail. Studies on integrating different hole transport materials (HTMs) and engineering the interfaces within the solar cell structure are expected to improve performance and pave the way for the commercial viability of lead-free rudorffite solar cells. Building on this research direction, Lee et al. have explored rudorffite structures, specifically Ag_3BiI_6 , as a promising lead-free alternative for solar cell applications.^[144] This approach aims to address the toxicity concerns associated with traditional lead-based perovskites while maintaining desirable optoelectronic properties. Bismuth (Bi) is used in place of lead due to its similar electronic configuration, which allows it to form stable and efficient perovskite structures with reduced toxicity and enhanced resistance to oxidation.

The architecture of the Ag_3BiI_6 rudorffite solar cell, depicted in Figure 5a, includes a layered structure comprising FTO, TiO_2 as the ETL, the Ag_3BiI_6 rudorffite absorber, Spiro-OMeTAD as the HTL, and an Ag electrode. This configuration is designed to facilitate efficient charge separation and minimize recombination losses. The UPS analysis in Figure 5b shows a binding energy difference of 0.70 eV between the VBM and the Fermi level, indicating a suitable energy alignment for photovoltaic applications.

The *J-V* characteristics of the solar cells, presented in Figure 5c, reveal those devices with Li-TFSI doped HTL achieve a PCE of 2.77%. This improvement underscores the critical role of HTL doping in enhancing device efficiency. Stability testing of unencapsulated Ag_3BiI_6 devices in atmospheric conditions, as shown in Figure 5d, demonstrates that devices made using the blade coating method retain $\approx 90\%$ of their initial PCE after 3000 h. In contrast, spin-coated devices exhibit a significant decline in PCE over time. The superior stability of blade-coated devices is attributed to their improved crystallinity and larger grain size, which reduce defect density and enhance resistance to moisture.

Sui et al. conducted an in-depth study on enhancing $Cs_2AgBiBr_6$ perovskite solar cells through hydrogenation treatment.^[145] Cesium (Cs) is selected for the A-site in $Cs_2AgBiBr_6$ perovskites because of its appropriate ionic size, which fits well into the perovskite lattice, providing stability and robustness. Cs-based perovskites offer better thermal and moisture stability compared to organic counterparts, crucial for the durability and longevity of solar cells. Cesium also aids in forming a stable perovskite phase at room temperature, essential for maintaining structural integrity and consistent performance over time.^[146]

As shown in Figure 5e, the preparation process starts with spin-coating a precursor solution of $Cs_2AgBiBr_6$ and DMSO onto a substrate, followed by applying an anti-solvent (IPA), then annealing the film at 290 °C. The final step involves hydrogen plasma treatment, which aims to incorporate atomic hydrogen into the perovskite lattice to improve its optoelectronic properties. This process integrates hydrogen atoms into the perovskite

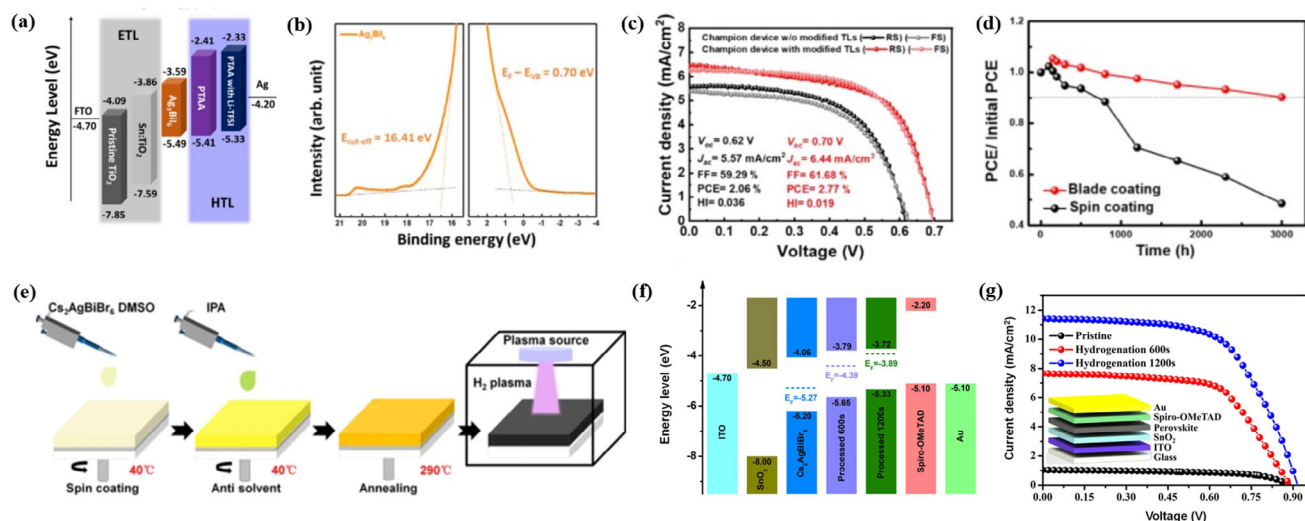


Figure 5. Lead-free wide-bandgap perovskite solar cell. a) Energy band alignment for Ag_4BiBr_6 perovskite solar cell and b) band structure analysis of Ag_3BiBr_6 layers using UPS. c) J - V curve of the solar cell with Li-TFSI doped HTL, and d) PCE of unencapsulated Ag_3BiBr_6 devices in atmospheric conditions. a-d) are reprinted with permission from,^[144] copyright 2023 Science Direct. e) Preparation method of $\text{Cs}_2\text{AgBiBr}_6$ perovskite films using hydrogen gas plasma hydrogenation treatment. f) Energy band alignment for $\text{Cs}_2\text{AgBiBr}_6$ PSC with different hydrogenation time (0, 600, and 1200 s). g) J - V curve for $\text{Cs}_2\text{AgBiBr}_6$ PSCs with different hydrogenation time (0, 600, and 1200 s). e-g) are reprinted with permission from,^[145] copyright 2022 Nature.

lattice, passivating defects and minimizing non-radiative recombination centers. The plasma treatment helps heal surface defects and grain boundaries, thereby enhancing the crystallinity and overall quality of the perovskite film.

Figure 5f shows the energy band alignment for $\text{Cs}_2\text{AgBiBr}_6$ solar cells with varying hydrogenation times. Initially, the pristine $\text{Cs}_2\text{AgBiBr}_6$ has a bandgap of 2.18 eV. This bandgap decreases to 1.91 eV after 600 s of hydrogenation and further to 1.64 eV after 1200 s. The reduction in the bandgap improves visible light absorption, boosting the overall efficiency of the solar cells. The J - V curves in Figure 5g illustrate the significant impact of hydrogenation on device performance. Solar cells treated with hydrogen for 1200 s achieved a peak PCE of 6.37%, compared to 5.64% for those treated for 600 s, and much lower efficiency for untreated samples.

7. Applications

PSCs have revolutionized the field of photovoltaics due to their impressive power conversion efficiencies, customizable bandgaps, and potential for economical production. These attributes have broadened the scope of perovskites beyond traditional solar panels, leading to breakthroughs in indoor photovoltaics (IPVs), tandem solar cells, and light-emitting applications.^[147,148] Their ability to enhance the light absorption spectrum of silicon makes perovskites ideal for tandem solar cells, achieving efficiencies beyond those of single-junction cells.^[149] Additionally, their excellent performance in low-light conditions makes them perfect for indoor photovoltaics, providing power for IoT devices and sensors in smart homes and industrial settings.^[150,151] Additionally, the flexibility of perovskite materials opens up new possibilities for wearable electronics, enabling integration into clothing and portable devices for continuous power generation and data collection.^[152]

Tandem solar cells have garnered considerable attention in the field of photovoltaics due to their potential to exceed the efficiency limitations of single-junction cells.^[30] The core concept of tandem solar cells involves stacking multiple layers of solar cells with different bandgaps, enabling more efficient utilization of the solar spectrum. In this configuration, high-energy photons are absorbed by the top cell with a wide bandgap, generating a high voltage, while lower-energy photons pass through to the bottom cell with a narrower bandgap, where they can still generate electricity.^[153] This layered approach maximizes the conversion of sunlight into electricity, leading to higher overall efficiencies. Early research primarily focused on silicon-based tandems, but the introduction of perovskite materials has dramatically transformed the field.^[154,155] Perovskites are particularly well-suited for tandem applications due to their high absorption coefficients, tunable bandgaps, and excellent charge-carrier properties. Recent advancements have seen perovskite-perovskite tandems achieving efficiencies over 26%, and perovskite-silicon tandems exceeding 30%, approaching the theoretical efficiency limits.^[156,157]

A key innovation in these advancements is the use of wide bandgap perovskites in the top cell of tandem configurations. Wide bandgap perovskites are ideal for effectively absorbing the high-energy portion of the solar spectrum. Their tunable bandgap allows for optimization when paired with silicon or other narrow-bandgap materials, ensuring each layer of the tandem structure operates at peak efficiency.^[158] Furthermore, the band structure of wide bandgap perovskites helps reduce thermalization losses, which occur when excess photon energy is dissipated as heat. By minimizing these losses, higher V_{OC} and overall improved efficiency are achieved.^[159] Moreover, wide bandgap perovskites enhance thermal stability and reduce recombination losses in tandem solar cells. Their ability to form ideal energy band alignments with various materials in the tandem setup

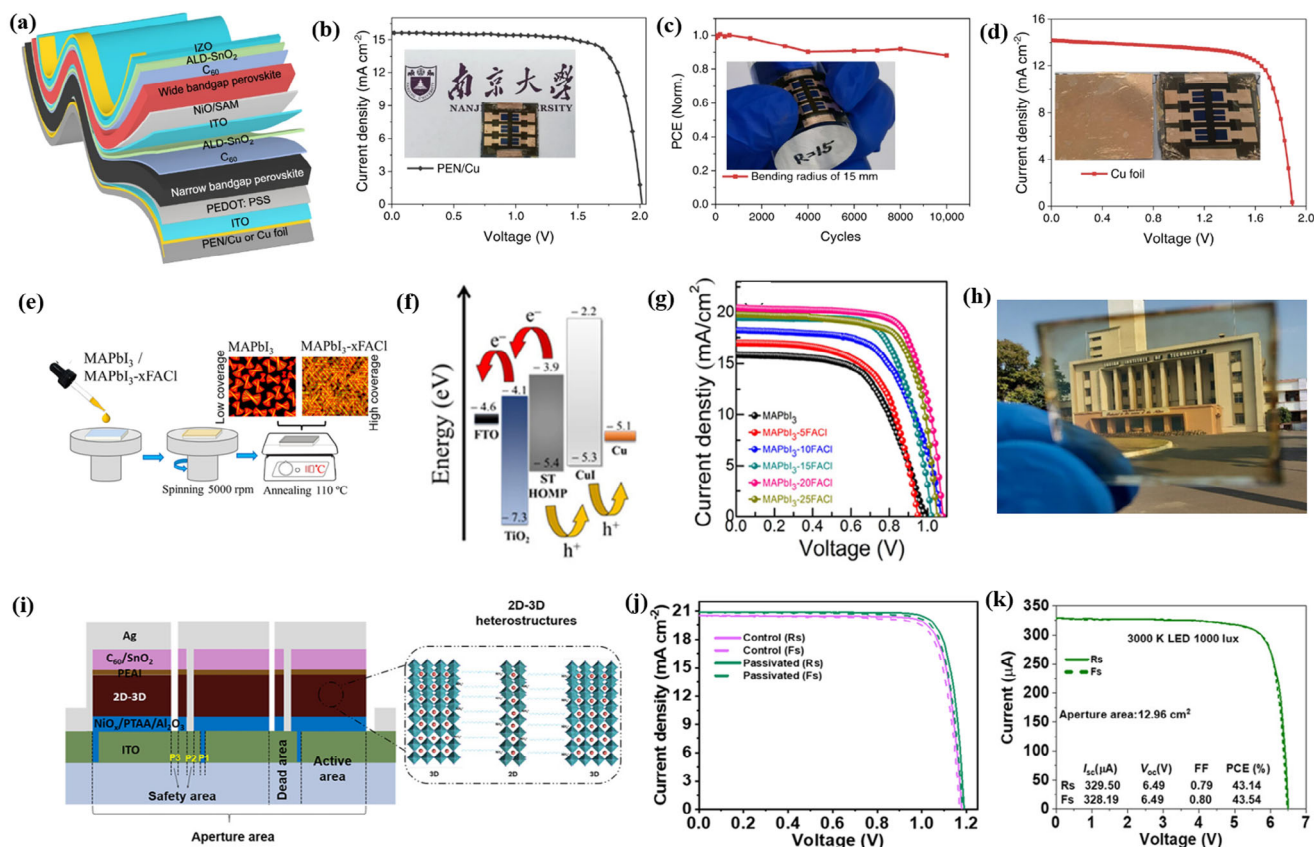


Figure 6. Application of wide bandgap perovskite. a) Device structure of a flexible substrate-configured tandem. b) J - V curves for flexible substrate-configured tandem using PEN. c) Bending tests of tandem cells on flexible substrates with a bending radius of 15 mm. d) J - V curves for flexible substrate-configured tandem using Cu-foils. a–d) are reprinted with permission from,^[161] copyright 2023 Nature. e) Schematic illustration of the fabrication process for MAPbI₃ and MAPbI_{3-x}FACl-based perovskite films. f) Energy band alignment for Semitransparent PSC. g) J - V curves as a function of FACl content. h) Digital photograph of the fabricated semitransparent PSC device demonstrating its semitransparency. e–h) are reprinted with permission from,^[176] copyright 2023 American Chemical Society. i) The schematic configuration of the fabricated semitransparent PSC device demonstrating its semitransparency. j) I - V curves of passivated (Mp) perovskite min-modules under AM1.5G illumination (100 mW cm⁻²) in reverse and forward scans. k) I - V curves of passivated (Mp) perovskite min-modules under LED illumination (1000 lux, 3000 K) in reverse and forward scans. i–k) are reprinted with permission from,^[184] copyright 2023 Science Direct.

ensures efficient charge separation and transport, which are crucial for high-performance devices.^[160]

Building on these principles, Tan et al. focus on developing flexible, high-efficiency tandem PSCs using different substrate materials.^[161] The device structure, depicted in Figure 6a, involves a tandem configuration with both wide-bandgap (WBG) and narrow-bandgap (NBG) perovskite layers. The layers are sequentially deposited on a flexible substrate, enhancing the cell's overall efficiency and mechanical durability. The fabrication process starts with spin-coating the perovskite layers on a flexible substrate, followed by thermal annealing to form high-quality perovskite films. The substrate materials used are polyethylene naphthalate (PEN) and copper (Cu) foils, each offering distinct advantages in terms of flexibility and electrical conductivity. PEN/Cu substrates provide a balance between flexibility and durability, while Cu foils offer superior electrical properties. Figure 6b shows the J - V curves for the tandem cells fabricated on PEN/Cu substrates. These devices achieved a PCE of 24.1%, demonstrating the effectiveness of PEN/Cu sub-

strates in enhancing device performance. The flexibility of PEN ensures the device can withstand mechanical stress while maintaining high efficiency. Mechanical durability tests, as shown in Figure 6c, highlight the robustness of these flexible tandem cells. The devices were subjected to bending tests with a bending radius of 15 mm, and their performance was monitored over 10,000 cycles. The results indicate that tandem cells on PEN/Cu substrates retained over 88% of their initial PCE, underscoring their potential for flexible and portable applications. Figure 6d presents the J - V curves for tandem cells on Cu foil substrates, achieving a PCE of 20.3%. The Cu foil's excellent electrical conductivity contributes to the overall performance, although the flexibility is slightly compromised compared to PEN.

Hybrid organic metal halide perovskite-based semitransparent solar cells have garnered significant attention for applications in smart windows, tandem devices, wearable electronics, displays, and sustainable Internet of Things (IoT) applications.^[162] Semitransparent perovskite solar cells take advantage of their unique optical and electronic properties to transmit sunlight while

efficiently converting it into energy. These cells achieve transparency through methods such as using thin perovskite films, applying dewetting techniques to form perovskite islands, employing self-assembly processes, and integrating transparent conductive layers.^[163–165] The tunable band gap of perovskite materials, which ranges between 1.5 and 2.3 eV depending on composition, plays a key role in optimizing both transparency and energy harvesting.^[166] A key feature of these solar cells is the transparent electrode, which facilitates light transmission while enabling effective charge extraction. Commonly used materials for transparent contacts include indium tin oxide (ITO), fluorine-doped tin oxide (FTO), silver nanowires (Ag NWs), and dielectric-metal-dielectric (DMD) structures.^[167–170] These components minimize optical losses while preserving conductivity. For example, DMD electrodes, which consist of a thin metallic layer (such as gold) positioned between dielectric layers, improve transparency and reduce reflection. Advances such as sputtered oxide coatings or solution-based nanowire networks further enhance both the efficiency and scalability of these technologies.^[171] These solar cells are highly suitable for building-integrated photovoltaic (BIPV) systems, serving as windows that allow light transmission while generating electricity. Adjusting the perovskite composition, such as the bromide-to-iodide ratio or the film thickness, enables a balance between transparency and PCE. Current advancements have achieved PCE values exceeding 10%, with visible light transmittance (AVT) ranging from 20% to 50%, while leveraging the material's band gap tunability to control light absorption.^[172–174] Looking forward, semitransparent perovskite solar cells are expected to enable scalable manufacturing and broad adoption. Research efforts are focusing on enhancing environmental durability while maintaining optimal transparency and efficiency. By utilizing their tunable band gap and multifunctionality, these solar cells have the potential to revolutionize energy solutions for buildings, allowing windows to act as power-generating components, contributing to sustainable energy generation, and reduced reliance on traditional energy sources.^[175]

In this study, Pradhan et al. introduce a single-step deposition method for formamidinium chloride (FACl)-mediated MAPbI₃ thin films.^[176] This method eliminates the need for an inert atmosphere and uses copper iodide (CuI) as the inorganic hole transport material to enhance the efficiency of transparent perovskite solar cells. The amount of FACl significantly influences the crystallinity, growth orientation, and internal strain of MAPbI₃ films, leading to improved charge carrier mobility and overall solar cell efficiency. By incorporating 20 mg mL⁻¹ of FACl into MAPbI₃, a PCE of 16.01% was achieved. Density functional theory (DFT) simulations provided insights into how FACl incorporation modifies the structural, electronic, and optical properties of MAPbI₃.

The fabrication process, illustrated in Figure 6e, involves spin-coating the perovskite layers on a substrate at 5000 rpm, followed by annealing at 110°C. This method enhances the coverage and quality of the MAPbI₃-xFACl films compared to pure MAPbI₃ films. Specifically, the addition of FACl leads to a more uniform film formation with higher coverage, as seen in the SEM images showing low and high coverage of MAPbI₃ and MAPbI₃-xFACl films, respectively. The energy band alignment for the semitransparent PSCs, shown in Figure 6f, demonstrates

efficient charge separation and transport, which are crucial for high-performance devices. The diagram indicates that the conduction band minimum and valence band maximum are well-aligned with the electron transport layer (TiO₂) and the hole transport layer (CuI), ensuring efficient charge extraction. The *J*-*V* curves in Figure 6g indicate that varying the FACl content directly impacts the performance of the PSCs. Specifically, the optimized MAPbI₃-20FACl composition achieves the highest PCE of 16.01%, demonstrating the importance of fine-tuning the FACl concentration for optimal device performance. The digital photograph in Figure 6h showcases the semitransparency of the fabricated device, which is a desirable feature for applications in smart windows and other integrated photovoltaic systems. The photograph clearly shows a building viewed through the semitransparent solar cell, illustrating its potential for integrating energy harvesting with transparency in various applications. Overall, the addition of FACl improves the crystal size and crystallinity of MAPbI₃ films, enhances charge mobility through strain modulation, and leads to the development of high-efficiency, semitransparent perovskite solar cells. The use of FACl provides unique advantages in crystal growth and passivation, leading to long-term stability and high performance in transparent solar cell applications.

Indoor PSCs are emerging as an effective solution for harnessing energy in indoor settings, where light intensity is lower, and the spectral distribution differs from that of outdoor light.^[177] These indoor PSCs are designed to efficiently capture energy from artificial light sources, such as LEDs, fluorescent lamps, and incandescent bulbs, commonly found in homes, offices, and industrial environments.^[178,179] The unique optoelectronic properties of perovskite materials, including their high light absorption and adjustable bandgap, make them particularly suitable for indoor photovoltaic applications.^[180]

The intensity, spectrum, and composition of indoor light sources differ significantly from sunlight. Indoor light intensity typically ranges from 200 lux in living rooms to 300–2000 lux in other environments, with artificial sources like fluorescent lamps, incandescent bulbs, halogen lamps, and LEDs emitting wavelengths between 380 and 780 nm. Unlike traditional solar cells designed for sunlight with narrow bandgaps, indoor photovoltaics (IPVs) utilize wider bandgaps (≈1.9 eV) to optimize performance under low-intensity light. Although indoor light produces lower current densities (μA cm⁻² compared to mA cm⁻² under AM 1.5G), this reflects reduced photon flux and does not indicate lower efficiency.^[181] Therefore, by tuning the bandgap of perovskite materials, the absorption spectrum can be matched to the emission spectrum of indoor light sources, enhancing energy conversion efficiency under low-light conditions.^[182] As the bandgap exceeds 1.9 eV, traditional designs based on *n*-type and *p*-type architectures may need to be adjusted or redesigned, requiring additional design efforts to accommodate wider bandgaps, ensuring effective charge transport and optimized energy conversion under indoor lighting conditions.^[183]

Mai et al. have explored the fabrication and performance of PSCs tailored for indoor applications, as shown in the accompanying figures.^[184] The fabrication process, depicted in Figure 6i, involves spin-coating the perovskite layers onto a substrate, followed by thermal annealing. This method produces high-quality

Table 1. PCE of wide bandgap Perovskite and Perovskite-like solar cells.

Composition	PCE	Bandgap	Strategy	Structure	Stability	Year/Ref.
MAPbBr ₂	12.08%	2.02 eV	Passivation	nip	—	2023 ^[185]
(FA _{0.83} CS _{0.17})Pb(I _{0.6} Br _{0.4}) ₃	17.0%	1.8 eV	Passivation	pin	25 °C, RH 50% 17 days 60%	2023 ^[77]
CS _{0.16} FA _{0.80} MA _{0.04} Pb(I _{0.68} Br _{0.32}) ₃	20.10%	1.75 eV	Passivation	pin	RH 30–50% 500h 91%	2023 ^[29]
(FA _{0.83} CS _{0.17})Pb(I _{0.6} Br _{0.4}) ₃	17%	1.79 eV	Passivation	pin	—	2022 ^[186]
CS _{0.3} FA _{0.6} DMA _{0.1} Pb(I _{0.7} Br _{0.3}) ₃	20.3%	1.75 eV	Deposition method	pin	25 °C 0.8sun 2560h ≈99%	2022 ^[39]
FA _{0.8} CS _{0.2} PbI _{2.4} Br _{0.6}	22.6%	1.67 eV	Passivation	nip	30 ± 5 °C 1000 h ≈81%	2022 ^[187]
FA _{0.5} CS _{0.5} PbI ₃	18.52%	1.65 eV	Additive	pin	RH 30%, 1000h ≈96%	2023 ^[188]
FA _{0.6} CS _{0.4} PbI ₃	17.69%	1.63 eV	Additive	nip	RH 20–25% 1000h 92%	2023 ^[189]
CS _{0.2} FA _{0.8} Pb(I _{0.7} Br _{0.3}) ₃	19.07%	1.75 eV	2D/3D	nip	RH 30% 1000h ≈85%	2022 ^[190]
CS _{0.2} FA _{0.8} Pb(I _{0.8} Br _{0.2}) ₃	20.11%	1.66 eV	Passivation	nip	N ₂ 1750h ≈85%	2023 ^[191]
CS ₂ PbI ₃	21.41%	1.71 eV	Passivation	nip	25 °C, RH 25% 900h ≈92.6%	2023 ^[192]
CS ₂ PbI ₃	17.8%	1.73 eV	Passivation	nip	30 °C, RH 30%, over 30 days ≈90%	2023 ^[193]
CS ₂ PbI ₂ Br	17.33%	1.89 eV	Photoactive layer	nip	RH < 10% 1000h ≈96.3%	2023 ^[88]
CS ₂ PbI ₂ Br ₂	9.38%	2.08 eV	Doped	nip	85 °C, RH 30%, 144h, ≈70%	2023 ^[89]
CS ₂ PbBr ₃	10.21%	—	HTL passivation	nip	85 % RH at 85 °C 30 days, 93%	2023 ^[194]
EDA _{0.01} (GA _{0.06} (FA _{0.8} CS _{0.2}) _{0.94}) _{0.98} SnI ₂ Br	8.66%	1.62 eV	Doped	pin	MPPT 200m ≈97.5%	2022 ^[124]
EDA(FAMA)Sn perovskite	11.03%	1.63 eV	Passivation	pin	N ₂ 1000h ≈80%	2023 ^[195]
PEA _{0.15} FA _{0.75} MA _{0.10} SnI ₂ Br	12.22%	1.65 eV	Additive	pin	Air 1200h ≈80%	2024 ^[196]
FA _{0.55} MA _{0.25} CS _{0.1} PMA _{0.1} SnBr ₂ I	6.2%	1.93 eV	Passivation	pin	N ₂ 2000h ≈95%	2024 ^[197]
Ag ₃ BiI ₆	0.70%	1.72 eV	Doped	nip	—	2022 ^[198]
AgBiI ₄	1.04%	1.81 eV	Additive	nip	—	2024 ^[199]
AgBiI ₄	1.87%	1.83 eV	Passivation	nip	25–30 °C RH 30–40% 16 days 95%	2024 ^[200]
Cu ₆ AgBiI ₁₀	0.041%	1.70–1.71 eV	—	nip	—	2023 ^[201]
(PEA) ₄ AgBiBr ₈	2.03%	2.8 eV	2D/3D	nip	—	2022 ^[202]
Cu _{0.4} Ag _{0.6} Sb _{0.95} Bi _{0.05} I ₄	0.7%	2.00–2.02 eV	Doped	nip	—	2023 ^[203]
CS ₂ Ag _{0.95} Ga _{0.05} BiBr ₆	4.52%	1.86 eV	Doped	nip	—	2024 ^[204]

perovskite films with excellent coverage and crystallinity. The 2D-3D heterostructures incorporating alkylamine oleylammonium ligands within the PSCs, illustrated in Figure 6i, is crucial for improving device performance by minimizing defects such as halide phase segregation and halide vacancy defects, which negatively impact the efficiency and stability of wide-bandgap perovskites. Figure 6j that, compared to the control device, the OAm⁺ ligand-treated cell achieves an improved PCE of 20.79%, which is higher than the control's 19.71%, demonstrating enhanced performance. The *I*–*V* curves in Figure 6k demonstrate the effect of different processing conditions on PSC performance. The best-performing device achieved a PCE of 43.14% under LED illumination (1000 lux, 3000 K), highlighting the potential of these devices for indoor energy harvesting applications. The device shows lower current under 1000 lux (μA range) compared to AM 1.5G (mA range) due to reduced photon flux, yet maintains high PCE in both conditions, highlighting its efficiency and adaptability to varying light intensities. This high efficiency is due to the optimized bandgap of the perovskite material, which improves light absorption and reduces thermalization losses. The *J*–*V* curve also shows that the optimized device exhibits high *V*_{OC} and *FF*, which are essential parameters to achieve high PCE. Their ability to maintain high efficiency under low-light conditions and their me-

chanical durability further enhance their suitability for indoor applications.

In summary, the development of indoor PSCs with optimized perovskite materials and advanced fabrication techniques offers significant promise for sustainable energy solutions in indoor environments. By leveraging the distinct properties of perovskites, these devices can efficiently convert indoor light into electrical energy, powering a wide range of electronic devices and contributing to the energy efficiency of modern indoor spaces. Ongoing research and innovation are crucial to overcoming existing challenges and fully realizing the potential of indoor perovskite solar cells (Table 1).

8. Conclusion and Outlook

Wide-bandgap (WBG) perovskite solar cells (PSCs) have recently demonstrated substantial potential to exceed the efficiency limits of single-junction solar cells, particularly the Shockley–Queisser limit. This detailed review focuses on the essential developments in WBG PSCs, highlighting significant innovations in material compositions, halide substitution techniques, and advanced device architectures that have propelled the field forward. One of the most impactful advancements has been the strategic substitution of iodine with bromine and the replacement of organic ions like

FA and MA with cesium. These changes have been crucial in attaining higher open-circuit voltages and minimizing thermalization losses, which are critical for boosting overall efficiency. Additionally, progress in integrating advanced charge transport layers and refining interface engineering techniques has played a vital role in reducing V_{OC} deficits and enhancing the photo-stability of WBG PSCs.

WBG PSCs have shown remarkable versatility, finding applications in semitransparent devices, indoor photovoltaics, and multijunction tandem solar cells. Their tunable bandgap allows for optimization tailored to specific applications, making them highly valuable. In tandem solar cells, for example, WBG PSCs can be paired with narrow-bandgap materials to maximize efficiency by utilizing a broader range of the solar spectrum.

However, despite these advancements, several challenges remain. Issues such as phase stability, ion migration, and long-term degradation under operational conditions need to be addressed to fully unlock the potential of WBG PSCs. Ongoing research is essential to improve material formulations, develop advanced encapsulation techniques, and design novel interfaces and charge transport layers that can mitigate these issues. Looking forward, the future of WBG perovskite solar cell technology hinges on overcoming these current limitations to produce highly efficient and stable devices that are viable for commercial photovoltaic applications. Key research priorities include enhancing stability through the development of more robust perovskite compositions and encapsulation methods, exploring new materials that offer better environmental stability and reduced toxicity (such as lead-free alternatives), optimizing device architectures and processing techniques to enhance performance, and tackling the challenges associated with scaling up production while maintaining cost-effectiveness and high performance. Expanding the application range of WBG PSCs beyond traditional uses to include tandem devices, indoor photovoltaics, and light-emitting applications is also crucial.

Interdisciplinary collaboration and continued innovation in materials science, engineering, and photovoltaics will be vital for realizing the full potential of WBG perovskite solar cells. Such efforts will not only advance photovoltaic technology but also contribute significantly to sustainable and efficient energy solutions for the future. By addressing the existing challenges and leveraging the unique properties of WBG PSCs, the development of next-generation solar cells that are both highly efficient and environmentally friendly can be achieved, leading to a more sustainable energy landscape.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

wide bandgap perovskite, bandgap engineering, compositional engineering, 2D perovskite, 3D perovskite

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